

SevaLens

NPU-Powered Offline Multilingual Document Assistant for Snapdragon PCs

Snapdragon® AI Lab Build & Present Challenge | Participant: Syed Ahmed | Individual entry

1. Problem and users

Online service (eSeva) shops process scholarship, certificate and welfare applications all day. Operators retype the same identity details into form after form, from scans and photos that mix English, Hindi, Urdu and Telugu. Cloud OCR tools would require uploading citizens' most sensitive documents. The users are shop operators and the citizens they serve.

2. Solution

SevaLens is a desktop assistant that runs entirely on the laptop. The operator scans or photographs a document; SevaLens reads it with on-device OCR, extracts and validates key fields (name, date of birth, ID number, address), and autofills the target application form for the operator to confirm. No document or extracted data leaves the machine.

3. AI models and stack

- **OCR:** a multilingual text-recognition model from Qualcomm AI Hub or an open-source platform, quantized for the NPU.
- **Field extraction:** a compact model plus rule-based validation for ID formats and confidence checks.
- **Runtime:** ONNX Runtime with the QNN execution provider targeting the Hexagon NPU; OpenCV for preprocessing.
- **Build:** native ARM64 Python for Windows on Snapdragon.

Final model selections will be confirmed against the current AI Hub catalog and documented in the repository.

4. Optimization for Snapdragon-powered PCs

Platform constraint	Risk for an AI app	SevaLens design response
Sustained thermal headroom	Long CPU-bound inference heats thin laptops and forces clock throttling	Run OCR and extraction on the Hexagon NPU with quantized models; CPU stays near idle
x86 emulation overhead	Translated x86/x64 code adds latency and battery cost	Native ARM64 Python and ONNX Runtime with the QNN execution provider; no emulated dependencies
Peak-power CPU spikes	Short high-clock bursts drain battery on everyday tasks	Batch pages onto the NPU; keep CPU work to light I/O and the UI

5. Performance targets and evaluation plan

Design targets, to be validated on Snapdragon hardware (not yet measured): latency of 1 second or less per page; 90% or more of inference operations on the NPU; CPU load under 15% during inference; no throttling over a 10-minute continuous run; OCR character error rate of 5% or less. Validation will compare CPU and NPU runs on the same 50-page sample of ID and form scans, using Qualcomm AI Hub profiling and on-device measurements, and results will be published in the repository README.

6. Deployment and accessibility

SevaLens ships as a single ARM64 install that works without internet, so it suits shops with unreliable connectivity. The interface supports the four languages above and keeps the operator in control of every autofilled value.

7. Innovation and roadmap

The novelty is a privacy-first, NPU-native workflow built from real service-shop practice rather than a generic OCR demo. Next steps: more form templates, Hindi/Urdu/Telugu voice input, and an audit log for operator corrections.